

Question:

Melanie is a door-to-door saleswoman. She sold a third of her vacuum cleaners at the green house, 2 more to the red house, and half of what was left at the orange house. If Melanie has 5 vacuum cleaners left, how many did she start with?

Reasoning + Answer:

<think>

Let x be total vacuum cleaners Melanie started with .

After Green house she had $x/3$ and at Red house she had $x/3 + 2$.

Half left at Orange house means $(x/3 + 2)/2$ and this is $x/3 + 2 - x/6 = 5$.

So $x/3 - x/6 + 2 = 5$, $x/6 + 2 = 5$, $x/6 = 3$, $x = 6 * 3 = 18$.

<answer> 18 </answer>

